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# MediSyn: A Synergistic multi-agent AI system for IPE in medical domain

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## Abstract

The increasing complexity of modern healthcare demands collaborative, inter-professional approaches to ensure comprehensive patient care, yet current AI solutions based on general large language models (LLMs) often fall short in delivering nuanced, profession-specific advice. This project addresses the challenge of integrating Inter-Professional Education/Practice (IPE/IPP) into an AI-driven multi-agent system designed to generate treatment plans for real-life patient cases. We demonstrate that a baseline single LLM solution performs poorly when tasked with generating treatment plans. To overcome these limitations, we propose an innovative approach that simulates real-world healthcare team interactions by incorporating a multi-agent system that dynamically creates agents based on the professions represented in each case study. Rather than relying on a hardcoded set of agents, our system extracts team member roles from real-world inter-professional case studies and assigns each agent the domain-specific knowledge and responsibilities of its corresponding profession. This approach allows, for example, a nurse agent to provide patient care insights and an oncologist agent to deliver targeted cancer treatment advice, closely mirroring real-world inter-professional collaboration. Our experiments show that this specialized, collaborative framework significantly outperforms the general-purpose LLM, paving the way for improved, context-aware treatment planning in clinical settings.

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# 1 Introduction

Interprofessional Education (IPE) is a pedagogical approach that brings together students from various healthcare disciplines to learn collaboratively, aiming to enhance teamwork and improve patient outcomes. This collaborative approach when put into practice is called Interprofessional Practice (IPP). Despite its recognized importance, implementing effective IPE within medical and public health education faces numerous challenges, including curriculum integration, scheduling conflicts, and resource limitations. Modern healthcare increasingly demands specialized, collaborative decision-making, yet current AI solutions based on general large language models (LLMs) often fall short in delivering nuanced, profession-specific advice.

In this project, we demonstrate that a baseline single LLM solution performs poorly when tasked with generating treatment plans. To overcome these limitations, we propose an innovative approach that simulates real-world healthcare team interactions by incorporating a multi-agent system that dynamically creates agents based on the professions represented in each case study. Rather than relying on a hardcoded set of agents, our system extracts team member roles from real-world inter-professional case studies and assigns each agent the domain-specific knowledge and responsibilities of its corresponding profession. This approach allows, for example, a nurse agent to provide patient care insights and an oncologist agent to deliver targeted cancer treatment advice, closely mirroring real-world inter-professional collaboration.

To address these issues, we propose the development of a synergistic multi-agent AI system designed to simulate educational conversations among diverse healthcare team members and generate treatment plans. By leveraging advanced technologies such as open-source Large Language Models (LLMs) and our novel agentic architecture, we compare its performance with a basic LLM response using evaluation metrics like BLEU scores, ROUGE scores and HITL methods.

*TODO: Our experiments show that this specialized, collaborative framework significantly outperforms general-purpose LLMs, paving the way for improved, context-aware treatment planning in clinical settings.*

## 2 Problem Statement

Effective interprofessional collaboration is essential for high-quality patient care, yet traditional educational methods struggle to adequately prepare students for real-world teamwork. Key challenges include difficulties in coordinating cross-disciplinary training, a lack of realistic simulations, and constraints on faculty resources.

Furthermore, existing AI-driven educational tools often fail to capture the nuanced decision-making processes that occur in interprofessional healthcare settings. Well firstly, there are hardly any AI solutions catered to the specific needs of IPE/IPP. Current AI systems in healthcare, predominantly powered by general large language models (LLMs), are designed to generate treatment plans in a one-size-fits-all manner. Although these models are capable of providing general answers, they lack the capacity to deliver nuanced, profession-specific insights essential for complex medical decision-making. This limitation is particularly evident when compared to real-world interprofessional collaboration, where specialized roles—such as those of nurses, oncologists, and pharmacists—contribute distinct expertise to patient care.

A single Large Language Model (LLM) approach lacks the specialization necessary to replicate the dynamic interplay between diverse healthcare professionals. This limitation leads to suboptimal educational

outcomes, as students do not experience the complexities of interprofessional collaboration in a meaningful way, or a patient won't be able to get a tailored treatment plan for their case.

To bridge this gap, our project proposes the development of a multi-agent AI system that can dynamically generate specialized virtual team members based on real-world case studies. This system aims to provide IPE/IPP practitioners and providers in the medical sector a tailored solution to generate more accurate and context-aware treatment plan for a patient's case, addressing the limitations of current methods and enhancing their readiness for clinical practice.

### 3 Related Work

The integration of artificial intelligence (AI) in healthcare has garnered significant attention in both academic research and industry applications. Early studies have focused on the deployment of general-purpose large language models (LLMs) to generate treatment plans; however, these models often fail to capture the nuanced expertise of specialized healthcare professionals, leading to suboptimal patient outcomes. Recent research highlights the importance of incorporating domain-specific knowledge into AI systems, particularly through multi-agent frameworks that better emulate interprofessional collaboration. For example, approaches such as the ClinicalAgent [1] framework employ distinct LLMs fine-tuned for specific clinical roles, showing improvements in diagnostic accuracy and treatment planning compared to monolithic models.

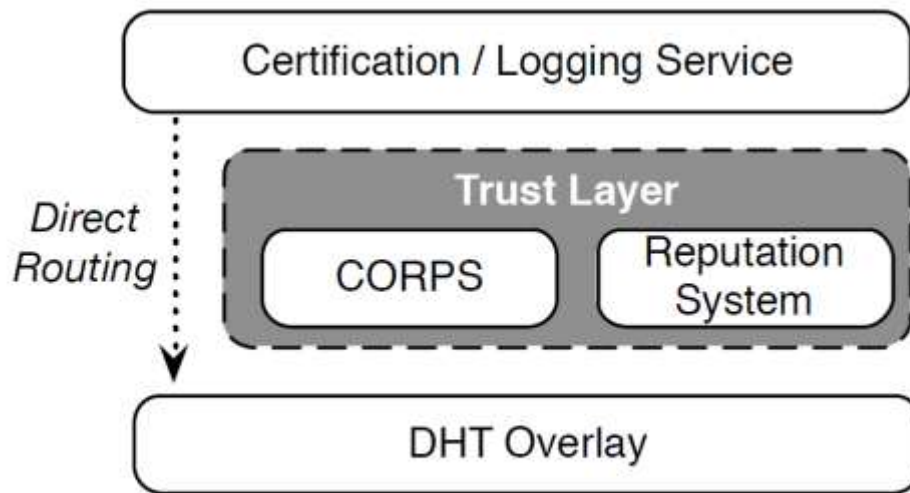
In parallel, multi-agent systems (MAS) have been extensively studied in healthcare for their ability to handle complex, distributed decision-making processes. MAS approaches have been applied to a range of tasks from emergency response and prehospital care management to cancer care information systems. These studies demonstrate that by decomposing healthcare tasks into specialized agents, systems can achieve more efficient and context-aware decision-making. Notably, the work by Mohammadzadeh et al. (2013) on cancer care information management [2] underscores how agents can process and exchange critical medical data to improve treatment outcomes

Your related work section positions your problem and your approach with respect to other, maybe similar, projects you've found in the literature. It "should not only explain what research others have done, but in each case should compare and contrast that to your work and also to other related work. After reading this section, a reader should understand the key idea and contribution of each significant piece of related work, how they \_t together, and how your work di\_ers." [3]<sup>1</sup> It's an edited and updated version of your literature review. Here are a few examples of how to insert citations like, [4] [3]and also [5], or even .

### 4 Solution and Methodology

The solution section covers all of your contributions (architecture, algorithms, formulas, findings). It explains in detail each contribution, if possible, with figures/schematics. Don't forget that a figure goes a long way towards helping your reader understand your work. For instance, Figure 1 outlines the layers involved in a distributed certification service, and how they articulate together. Nevertheless, a figure must always come with at least one paragraph of

explanation. The rule is that anyone should be able to understand your solution from reading the text in this section, even if they skip the figures.



*Figure 1: Architecture of our distributed certification service*

Figure 2 is a pretty good example of a figure that is completely useless unless it is not accompanied by a textual explanation.

## 5 Results and Discussion

The results section details your metrics and experiments for the assessment of your solution. It then provides experimental validation for your approach with visual aids such as data tables and

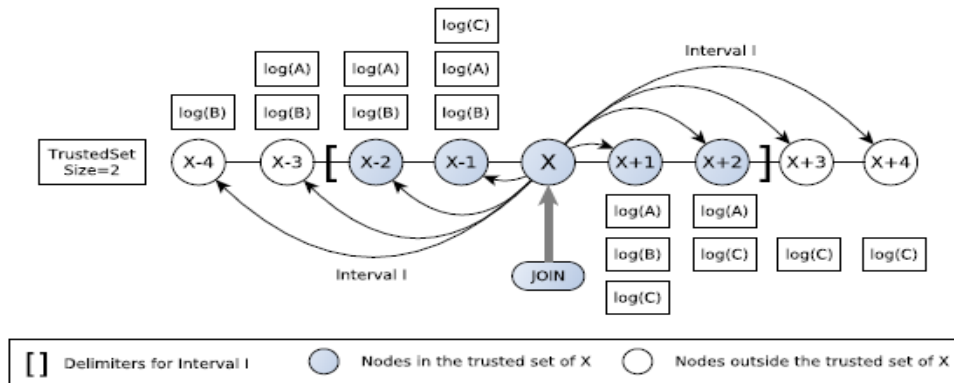


Figure 2: Try to guess what this gure illustrates; I double-dare you.

graphs. In particular, it allows you to compare your idea with other approaches you've tested, for example solutions you've mentioned in your related work section.

## 5.1 Experimentation protocol

It is of the utmost importance to describe how you came up with the measurements and results that support your evaluation.

## 5.2 Data tables

Every data table should be numbered, have a brief description as its title, and specify the units used.

As an example, Table 1 compares the average latencies of native application calls to networked services. The experiments were conducted on an Apple MacBook Air 2010 with a CPU speed of 1.4GHz and a bus speed of 800MHz. Each data point is a mean over 20 instances of each call, after discarding both the lowest and the highest measurement.

| Network Applications |          |              |
|----------------------|----------|--------------|
| Service              | Protocol | Latency (ms) |
| DNS                  | UDP      | 13.65 ms     |
|                      | TCP      | 0.01 ms      |
| NTP                  | UDP      | 92.50 ms     |
| SMTP                 | TCP      | 33.33 ms     |
| HTTP                 | TCP      | 8.99 ms      |

### 5.3 Graphs

Graphs are often the most important information in your report; you should design and plot them with great care. A graph contains a lot of information in a short space. Graphs should be numbered and have a title. Their axes should be labelled, with the quantities and units specified. Make sure that individual data points (your measurements) stand out clearly. And of course,

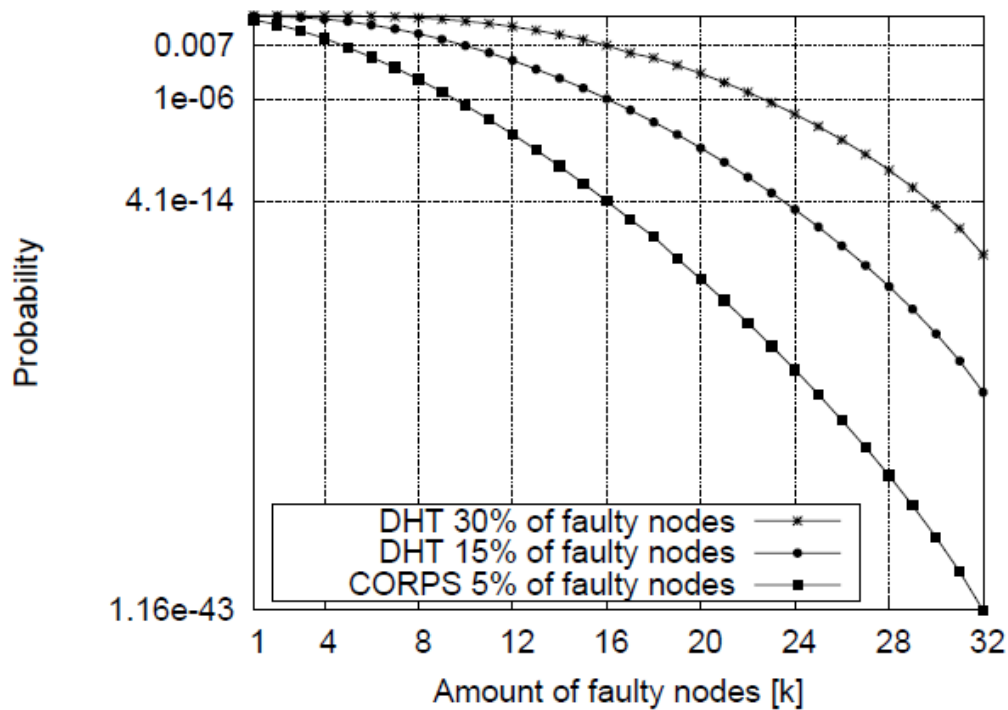


Figure 3: Probability of including  $[k]$  faulty/malicious nodes in the service

always associate your graph with text that explains your results, and outlines the conclusions you draw from these results.

For example, Figure 3 compares the efficiency of three different service architectures in eliminating adversarial behaviors. Every data point gives the probability that  $k$  faulty/malicious nodes managed to participate in a computation that involves 32 nodes. In the absence of at least one reliable node ( $k = 32$ ), the failure will go undetected; but the results show that this case is extremely unlikely, regardless of the architecture. The most significant result pertains to  $k = 16$ : the reliable nodes detect the failure, but cannot reach a majority to recover. The graph shows that the CORPS 5% architecture is much more resilient than the DHT 30% architecture, by a magnitude of  $10^{11}$ .

## 6 Discussion

The discussion section focuses on the main challenges/issues you had to overcome during the project. Outline what your approach does better than the ones you mentioned in your related work, and explain why. Do the same with issues where other solutions outperform your own. Are there limitations to your approach? If so, what would you recommend towards removing/mitigating them? Given the experience you've gathered working on this project, are there other approaches that you feel are worth exploring?

## 7 Conclusion

Give a clear, short, and informative summary of all your important results. Answer the initial question(s) or respond to what you wanted to do, as stated in your introduction. It can be a short table or a list, and possibly one or two short comments or explanations.

Target a reader who may not have time to read the whole report yet, but needs the results or the conclusions immediately. This is a typical situation in real life. Some readers will read your introduction and skip to your conclusion first, and read the whole report only later (if at all). You may also draw perspectives. What's missing? In what directions could your work be extended?

## 8 Bibliography

- [1] R. G. a. F. C. G. a. S. A. a. S. B. a. E. K. a. M. K. a. D. C. a. A. Kamen, "Multi-Agent Reinforcement Learning Meets Leaf Sequencing in Radiotherapy," in *ICML'24*, Vienna, Austria, 2024.
- [2] N. S. R. & R. A. Mohammadzadeh, "Mohammadzadeh N, Safdari R, Rahimi A. Multi-agent systems: effective approach for cancer care information management.," *Asian Pacific journal of cancer prevention : APJCP*, 2013.
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- [4] V. P. a. L. Iftode, "Byzantine fault tolerant public key authentication in peer-to-peer," *Computer Networks*, 2006..
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